

Title Observational Strategy of Ultra-short Period Exoplanets with Ariel

Institution Kapteyn Astronomical Institute

Participant(s)	Name	Email address	Role
S4220110	L. Cesario	l.c.cesario@student.rug.nl	Owner
P311056	T. Lichtenberg	tim.lichtenberg@rug.nl	Supervisor
P317904	Q. Changeat	q.changeat@rug.nl	Supervisor

StartDate	2025-09-01	EndDate	2026-07-01
Number	FSE-20260304-00231	Publish	2026-03-04

0.1 The reproducibility of results is vital to science. Aim of this RDMP is to have you think in advance how you will deal with data, data processing and simulations to make sure that your work done during the PhD or the Master project will be and remain reproducible. The RDMP should be discussed with your supervisor. The first version of your RDMP is required before the 6th month in a PhD and before the 3rd month of a master project. This version must be approved by the head of the computer group. The RDMP should be updated on a regular basis. The final version of the RDMP, describing data management has to be approved by the supervisor before the final grading or submission of the PhD manuscript. I agree:	Yes
1.0 Data Management Plan	
1.1 Title of the research project	Observational Strategy of Ultra-short Period Exoplanets with Ariel
1.10 If your thesis or RDMP do not fully describe results and/or origin of the data, then please enter	N/A

additional information.	
1.2 Supervisor(s)	Tim Lichtenberg, Quentin Changeat
1.3 What is the start date of your project?	2025-09-01
1.4 Give a brief (<200 words) description of your project	In my project I combine the long-term physical evolution of exoplanets using the PROTEUS framework with the Ariel mission simulator. The project entails obtaining the best (if possible) observational strategy for these planets for ESA's future mission Ariel assuming they have a thick volatile-rich atmosphere.
1.5 Will you be using existing measurement data?	Yes
1.5.1 Origin of the existing data: - Where can it be retrieved? - How can it be retrieved?	Can be retrieved on the NASA Exoplanet Archive by looking up the planet's names
1.6 Has the existing data that you will use, already been processed?	Yes
1.6.1 Give a brief description of the processed data : - Where is the data located? - How can it be retrieved?	By pressing on the download buttons in the Exoplanet Archive (NASA) for the specific planets.

1.7	Yes
Do you plan to process data yourself?	
1.7.1	
Give a brief description of how you will process (have processed) the data. • Which software that you will use (name and version). • Is this software for which a license is needed? • If it is software developed by you, where is it stored? • If you used a data disk in your workstation, give the name of the workstation and a path to the relevant folders. • If you compiled software then write a README with information how it was compiled. • If you compiled software, please add version information also.	I used the PROTEUS software to simulate, from the initial conditions data of these planets, their long-term evolution which obtains new data in the form of possible planetary evolution outcomes. The software is developed by my research group and is found on GitHub (fully accessible and reproduceable).
1.8	Yes
Will you be doing simulations?	
1.8.1	
Give a brief description on how you will perform (have performed) the simulations: • Software used • Input parameters • Describe data formats • Describe homebrew code and scripts	The software used is PROTEUS for the planet evolution. The input parameters are those obtained by NASA's archive, such as orbital configurations/mass/radius/star type/etc. The output of the simulations is updated planetary characteristics. Most notably their thermal profiles, chemical compositions and temperatures. My scripts work on processing this output data in other code frameworks such as TauREx for radiative transfer simulation and ArielRAD for convolution of spectra with instrument noise.
1.9	N/A

Thesis (to be filled in the final version of the RDMP):

Provide the location of all electronic versions of the tables, plots and images of the thesis.